- We would like to implement a one unit or one standard deviation increase in the error of one of the variables in the model (while holding other quantities unchanged) and then trace out the behaviors of other variables over time.

- Consider

$$\mathbf{y}_t = \boldsymbol{\mu} + \boldsymbol{\varepsilon}_t + \boldsymbol{\Psi}_1 \boldsymbol{\varepsilon}_{t-1} + \boldsymbol{\Psi}_2 \boldsymbol{\varepsilon}_{t-2} + \dots$$

  The impulse response of $\mathbf{y}_t$ with respect to a one-unit shock in ε_j (the jth element in $\boldsymbol{\varepsilon}$) at period h is given by $\boldsymbol{\Psi}_h \mathbf{e}_j$, where $\mathbf{e}_j$ is a $k \times 1$ vector of 0s with 1 in position j. The coefficient matrices of the MA representation of a VAR(p) model are known as the *impulse response functions* (IRF) of $\mathbf{y}_t$.

- A shock of size one may not always be economically meaningful. It is a common practice to normalize the size of the shocks to be one standard deviation. Under this convention, the impulse response of $\mathbf{y}_t$ to a one-standard-deviation of ε_j at period h is given by $\sigma_j \boldsymbol{\Psi}_h \mathbf{e}_j$, where σ_j is the standard deviation of ε_j.

- The jth column of the summation $\sum_{h=0}^{s} \boldsymbol{\Psi}_h$ contains the *accumulated responses* over s periods to a unit shock in the jth variable of the system. These quantities are also known as the jth *interim multipliers*. The total accumulated effects for all future periods $\sum_{h=0}^{\infty} \boldsymbol{\Psi}_h$ are called the *long-run effects* or *total multipliers*.

- The above analysis assumes that a shock occurs only to one variable at a time. This assumption may not be reasonable in empirical applications. Since the elements of $\boldsymbol{\varepsilon}_t$ are generally correlated (i.e., $\boldsymbol{\Sigma}$ is not a diagonal matrix), a change in one component of $\boldsymbol{\varepsilon}_t$ will simultaneously affect other components.

- To overcome this difficulty, we may want to choose a matrix $\mathbf{P}$ such that $\mathbf{v}_t \equiv \mathbf{P}^{-1} \boldsymbol{\varepsilon}_t$ are uncorrelated (*orthogonal*) and have unit variances, namely, $E(\mathbf{v}_t \mathbf{v}_t') = \mathbf{I}_k$.

- We can use the Cholesky decomposition to find the $\mathbf{P}$ (Sims, 1980):

$$\boldsymbol{\Sigma} = \mathbf{P}\mathbf{P}',$$

where $\mathbf{P}$ is a lower-triangular matrix. Then

$$E\left(\mathbf{v}_t\mathbf{v}_t'\right) = E\left(\mathbf{P}^{-1}\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t'\mathbf{P}^{-1\prime}\right) = \mathbf{P}^{-1}\boldsymbol{\Sigma}\mathbf{P}^{-1\prime} = \mathbf{I}_k.$$

- The moving average representation (12) can be rewritten as

$$\begin{aligned}\mathbf{y}_t &= \boldsymbol{\mu} + \boldsymbol{\varepsilon}_t + \boldsymbol{\Psi}_1\boldsymbol{\varepsilon}_{t-1} + \boldsymbol{\Psi}_2\boldsymbol{\varepsilon}_{t-2} + \ldots \\ &= \boldsymbol{\mu} + \mathbf{P}\mathbf{P}^{-1}\boldsymbol{\varepsilon}_t + \boldsymbol{\Psi}_1\mathbf{P}\mathbf{P}^{-1}\boldsymbol{\varepsilon}_{t-1} + \boldsymbol{\Psi}_2\mathbf{P}\mathbf{P}^{-1}\boldsymbol{\varepsilon}_{t-2} + \ldots \\ &= \boldsymbol{\mu} + \boldsymbol{\Psi}_0^*\mathbf{v}_t + \boldsymbol{\Psi}_1^*\mathbf{v}_{t-1} + \boldsymbol{\Psi}_2^*\mathbf{v}_{t-2} + \ldots, \end{aligned} \tag{13}$$

  where $\boldsymbol{\Psi}_0^* = \mathbf{P}$ and $\boldsymbol{\Psi}_i^* = \boldsymbol{\Psi}_i\mathbf{P}$, for $i = 1, 2, \ldots$. The coefficient matrices $\boldsymbol{\Psi}_i^*$ are called the *impulse response function* of $\mathbf{y}_t$ with respect to the *orthogonal innovations* $\mathbf{v}_t$, or simply *orthogonalized impulse response function* (OIRF). Specifically, the (ℓ, j)th element of $\boldsymbol{\Psi}_i^*$ is the impact of a shock with size one standard deviation of the jth innovation at time t on the future value of $y_{\ell,t+i}$. We can define accumulated responses in a similar manner as before.

- *Example*: Consider

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} 0.5 & 0.2 \\ 0.3 & 0.4 \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

  where $\boldsymbol{\varepsilon}_t \sim WN\left(\mathbf{0}, \boldsymbol{\Sigma}\right)$ with

$$\boldsymbol{\Sigma} = E\left(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t'\right) = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}.$$

  First, we can verify that this process is weakly stationary since

$$|\boldsymbol{\Phi} - \lambda\mathbf{I}_2| = det\left[\begin{pmatrix} 0.5 & 0.2 \\ 0.3 & 0.4 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right] = 0$$

has roots $\lambda_1 = 0.94$ and $\lambda_2 = -0.04$, both lying inside the unit circle. The MA representation is $\mathbf{y}_t = \sum_{i=0}^{\infty} \mathbf{\Psi}_i \boldsymbol{\varepsilon}_{t-i}$, with $\mathbf{\Psi}_0 = \mathbf{I}_2$, $\mathbf{\Psi}_1 = \mathbf{\Phi}$, and so on. The Cholesky decomposition of $\mathbf{\Sigma}$ gives

$$\mathbf{P} = \begin{pmatrix} 1.41 & 0 \\ 0.70 & 1.87 \end{pmatrix} \quad \text{and} \quad \mathbf{P}^{-1} = \begin{pmatrix} 0.70 & 0 \\ -0.27 & 0.53 \end{pmatrix}.$$

Then we can compute the orthogonalized impulse responses by

$$\begin{aligned} \mathbf{y}_t &= \mathbf{P}\mathbf{P}^{-1}\boldsymbol{\varepsilon}_t + \mathbf{\Psi}_1\mathbf{P}\mathbf{P}^{-1}\boldsymbol{\varepsilon}_{t-1} + \mathbf{\Psi}_2\mathbf{P}\mathbf{P}^{-1}\boldsymbol{\varepsilon}_{t-2} + \dots \\ &= \mathbf{P}\mathbf{v}_t + \mathbf{\Psi}_1\mathbf{P}\mathbf{v}_{t-1} + \mathbf{\Psi}_2\mathbf{P}\mathbf{v}_{t-2} + \dots \end{aligned}$$

or

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} 1.41 & 0 \\ 0.70 & 1.87 \end{pmatrix} \begin{pmatrix} v_{1t} \\ v_{2t} \end{pmatrix} + \begin{pmatrix} 0.85 & 0.37 \\ 0.70 & 0.75 \end{pmatrix} \begin{pmatrix} v_{1,t-1} \\ v_{2,t-1} \end{pmatrix} + \dots$$

Note that v_{2t} does not have instantaneous (or contemporaneous) effects on y_{1t}.

- A weakness of this approach is that this definition of impulse response function depends on the *ordering* of the components in $\mathbf{y}_t$. The lower triangular structure of $\mathbf{P}$ (and also $\mathbf{P}^{-1}$) indicates that v_{1t} is a function of ε_{1t}, v_{2t} is a function of ε_{1t} and ε_{2t}, and so on. In other words, v_{it} is not affected by ε_{jt} for $j > i$.

- In a three-variable VAR, this assumption implies that the first shock affects all of the variables instantaneously. The second shock only affects the second and third variables instantaneously. The third shock only affects the third variable instantaneously.

- More generally, y_{jt} does not have an instantaneous impact on $y_{\ell t}$ in the preceding equations ($\ell < j$). Thus, this method assumes a "causal chain" for the shocks.

- Different variable orderings in the VAR lead to different orthogonalized impulse response functions. The orthogonalized responses are invariant to the ordering of the variables only if $\mathbf{\Sigma}$ is diagonal.

- In practice, economic theory can provide guidance on the ordering. It may be plausible to argue that some variables are sticky and do not respond immediately to certain shocks.

- For example, Stock and Watson (2001) consider a three-variable VAR ordered as 1) inflation, 2) the unemployment rate, and 3) the interest rate. The identifying assumptions are

    1. Inflation depends only on lagged values of the other variables, possibly reflecting sticky prices.
    2. Unemployment responds to current inflation but not to the current funds rate.
    3. The federal funds rate responds to both current inflation and unemployment, consistent with the Fed using real-time economic information when setting interest rates.

- However, in many cases there is no clear *a priori* knowledge. Sims (1981) recommends trying different orderings and assessing how sensitive the orthogonalized impulse responses—and the resulting conclusions—are to the chosen ordering.

- Other methods have been developed to overcome the issue that orthogonalized impulse responses depend on the ordering of variables in a VAR. The leading alternative is the generalized impulse response function (GIRF) proposed by Koop et al. (1996):

$$GIRF\left(h,\delta_i,\mathbf{\Omega}_{t-1}\right)\equiv E\left(\mathbf{y}_{t+h}|\varepsilon_{it}=\delta_i,\mathbf{\Omega}_{t-1}\right)-E\left(\mathbf{y}_{t+h}|\mathbf{\Omega}_{t-1}\right).$$

  It measures the expected change in $\mathbf{y}_{t+h}$ due to a shock of size $\delta_i=\sigma_i$ in the ith equation, where $\mathbf{\Omega}_{t-1}$ is the information set at time $t-1$.

- In the context of VAR models, using the $MA\left(\infty\right)$ representation again, we obtain

$$GIRF\left(h,\delta_i,\mathbf{\Omega}_{t-1}\right)=\mathbf{\Psi}_h E\left(\boldsymbol{\varepsilon}_t|\varepsilon_{it}=\delta_i\right).$$

  The computation of $E\left(\boldsymbol{\varepsilon}_t|\varepsilon_{it}=\delta_i\right)$ depend on the multivariate distribution assumed for the disturbances $\boldsymbol{\varepsilon}_t$.

- If $\boldsymbol{\varepsilon}_t\sim IIDN\left(\mathbf{0},\mathbf{\Sigma}\right)$, then $E\left(\boldsymbol{\varepsilon}_t|\varepsilon_{it}=\delta_i\right)=\left(\sigma_{1i},\sigma_{2i},\ldots,\sigma_{ki}\right)'\sigma_{ii}^{-1}\delta_i$, where $\mathbf{\Sigma}=\left(\sigma_{ij}\right)$ and

$\sigma_{ii} = \sigma_i^2$. Therefore, for a shock of size $\delta_i = \sigma_i$,

$$GIRF\left(h, \delta_i, \mathbf{\Omega}_{t-1}\right) = \mathbf{\Psi}_h \mathbf{\Sigma} \mathbf{e}_i \sigma_i^{-1},$$

where $\mathbf{e}_i$ is a selection vector with 1 in the ith position and 0 elsewhere. The jth element of this expression is the GIRF of a one standard deviation shock to the ith equation on the jth variable at horizon h.

- The GIRF is invariant to variable ordering because it integrates the effects of other shocks out of the response.
- Inference for IRFs
  - Under standard regularity conditions, the estimated VAR coefficients are asymptotically normal. As a result, impulse responses are also asymptotically normal by the delta method.
  - However, analytical derivation of the IRF covariance matrix is tedious, and asymptotic approximations may be inaccurate since the IRFs are highly nonlinear functions of the coefficients.
  - In practice, bootstrap methods are often used to construct confidence intervals for IRFs.

2.8 Forecast Error Variance Decomposition

- To understand which shocks drive the variability of each variable in the system over time, we can decompose the forecast error variance of each variable into contributions from different structural shocks.
- Forecast error variance decomposition requires orthogonalized shocks. Using equation (13):

$$\mathbf{y}_t = \boldsymbol{\mu} + \mathbf{\Psi}_0^* \mathbf{v}_t + \mathbf{\Psi}_1^* \mathbf{v}_{t-1} + \mathbf{\Psi}_2^* \mathbf{v}_{t-2} + \ldots,$$

we obtain the error of the optimal h-step ahead forecast at the forecast origin t:

$$\mathbf{e}_{t+h|t} \equiv \mathbf{y}_{t+h} - \hat{\mathbf{y}}_{t+h|t} = \mathbf{\Psi}_0^* \mathbf{v}_{t+h} + \mathbf{\Psi}_1^* \mathbf{v}_{t+h-1} + \ldots + \mathbf{\Psi}_{h-1}^* \mathbf{v}_{t+1} = \sum_{\ell=0}^{h-1} \mathbf{\Psi}_\ell^* \mathbf{v}_{t+h-\ell}.$$

and then

$$Var\left(\mathbf{e}_{t+h|t}\right) = E\left(\mathbf{e}_{t+h|t}\mathbf{e}_{t+h|t}'\right) = \sum_{\ell=0}^{h-1}\sum_{j=0}^{h-1} \mathbf{\Psi}_\ell^* E\left(\mathbf{v}_{t+h-\ell}\mathbf{v}_{t+h-j}'\right)\mathbf{\Psi}_j^{*\prime} = \sum_{\ell=0}^{h-1} \mathbf{\Psi}_\ell^* \mathbf{\Psi}_\ell^{*\prime}, \quad (14)$$

where the last equality follows by $E\left(\mathbf{v}_t\mathbf{v}_t'\right) = \mathbf{I}_k$.

- Let $\mathbf{\Psi}_\ell^* = (\psi_{ij,\ell})$. It follows from (14) that the variance of the forecast error $e_{i,t+h|t}$, which is the ith element of $\mathbf{e}_{t+h|t}$, is given by

$$Var\left(e_{i,t+h|t}\right) = \sum_{\ell=0}^{h-1}\sum_{j=1}^{k} \psi_{ij,\ell}^2 = \sum_{j=1}^{k} \omega_{ij,h} \quad (15)$$

where

$$\omega_{ij,h} \equiv \sum_{\ell=0}^{h-1} \psi_{ij,\ell}^2.$$

Therefore, $\omega_{ij,h}$ can be interpreted as the contribution of the jth shock v_{jt} to the variance of the h-step ahead forecast error of y_{it}. The proportion of $Var\left(e_{i,t+h|t}\right)$ due to shock v_{jt} is $\omega_{ij,h}/Var\left(e_{i,t+h|t}\right)$. Equation (15) is called the (orthogonalized) *forecast error variance decomposition* (FEVD).

- The interpretation of the FEVD is subject to the same problem as the impulse response analysis – its decomposition is not unique as it depends on the choice of the transformation matrix $\mathbf{P}$.

3 Unit Root and Cointegration

3.1 Cointegration

- Many economic time series are nonstationary but exhibit long-run equilibrium relationships—cointegration helps us model and analyze these stable relationships despite short-term

fluctuations.

- Two or more variables are said to be *cointegrated* if each of them is unit root nonstationary (i.e., integrated of order one), but there exist linear combinations of them that are stationary.

- Intuitively, cointegration means that two or more series share a common stochastic trend, which can be eliminated by taking some specific difference of the series such that the resulting series is stationary.

- Consider $y_{1t}, y_{2t}, \ldots, y_{kt}$, where $y_{it} \sim I(1)$, $i = 1, 2, \ldots, k$. The vector $\mathbf{y}_t = (y_{1t}, y_{2t}, \ldots, y_{kt})'$ is said to be *cointegrated* if there exists at least a $k \times 1$ vector, $\boldsymbol{\beta}$, such that $\boldsymbol{\beta}'\mathbf{y}_t \sim I(0)$. $\boldsymbol{\beta}$ is called a *cointegrating* vector, and $\boldsymbol{\beta}'\mathbf{y}_t$ is known as the *long-run relation.*

- A cointegrating vector is not unique. Clearly, multiplying $\boldsymbol{\beta}$ by any nonzero constant yields another valid cointegrating vector. To uniquely identify $\boldsymbol{\beta}$, some normalization assumption is needed— typically by setting some component of $\boldsymbol{\beta}$ to one. Moreover, there may exist multiple linearly independent cointegrating vectors, each representing a distinct long-run equilibrium relationship.

- Note that in general, linear combinations of $I(1)$ series are $I(1)$. The absence of cointegration indicates that there is no long-run equilibrium relationship among the variables.

- Examples of plausible cointegrating relations (supported by economic theory):

  - Log consumption and log income (permanent income hypothesis): the consumption/income ratio tends to be stable over long periods of time, implying that (log consumption − log income) is stationary.
  - Log stock price and log dividend (present value model): the dividend/price ratio is stationary since a stock's price reflects the discounted present value of expected future dividends.
  - Nominal exchange rate and foreign and domestic prices (Purchasing Power Parity)

- Nominal interest rates of different maturities (expectations hypothesis of the term structure)
- Nominal interest rates and inflation (Fisher equation)
- Stock prices of a company on different stock exchanges (Law of One Price)

- Why do we care about cointegration?
  - In economics, cointegration characterizes a long-run equilibrium relationship. Long term forecasts of cointegrating series must be linearly related.
  - In finance, cointegration underpins the pairs trading strategy, which involves taking offsetting long and short positions in two stocks that are cointegrated.
  - Statistical reason:
    * Standard regression analysis fails when dealing with $I(1)$ variables, leading to *spurious regressions*, which show misleading statistical evidence of a linear relationship even when the series are actually independent of each other (Granger and Newbold, 1974).
    * OLS estimates of cointegrating vectors are consistent even when the regressors are correlated with the error term, and the estimates converge to the true values at a faster rate, T, rather than the usual $\sqrt{T}$ rate—this property is known as "super-consistency".

- *Example*: Consider the following system of processes

$$
\begin{aligned}
y_{1t} &= \gamma_1 y_{2t} + \gamma_2 y_{3t} + \varepsilon_{1t} \\
y_{2t} &= \gamma_3 y_{3t} + \varepsilon_{2t}, \\
y_{3t} &= y_{3,t-1} + \varepsilon_{3t},
\end{aligned}
$$

where $\varepsilon_{it} \sim WN\left(0, \sigma_i^2\right)$, for $i = 1, 2, 3$. It can be deduced that all the three processes are individually $I(1)$. Let $\mathbf{y}_t = (y_{1t}, y_{2t}, y_{3t})'$ and $\boldsymbol{\beta}_1 = (1, -\gamma_1, -\gamma_2)'$. Then $\boldsymbol{\beta}_1' \mathbf{y}_t = \varepsilon_{1t} \sim$

$I(0)$. Another cointegrating relationship, which only involves y_{2t} and y_{3t}, is $\boldsymbol{\beta}_2'\mathbf{y}_t = \varepsilon_{2t} \sim I(0)$, where $\boldsymbol{\beta}_2 = (0, 1, -\gamma_3)'$.

3.2 Residual-Based Tests for Cointegration

- Testing for cointegration can be viewed as testing for the existence of long-run equilibria among two or more series.
- Engle and Granger's (1987) two-step procedure:
  - Step 1: Form the cointegrating residual $\boldsymbol{\beta}'\mathbf{y}_t = \varepsilon_t$.
  - Step 2: Perform a unit root test on ε_t (using ADF, Philips-Perron, or other unit root tests). The null hypothesis is $H_0 : \varepsilon_t \sim I(1)$ (no-cointegration), and the alternative is $H_1 : \varepsilon_t \sim I(0)$ (cointegration).
- Two cases: Do we know the proposed cointegrating vector $\boldsymbol{\beta}$?

Case 1. Yes, $\boldsymbol{\beta}$ is pre-specified (not estimated), often based on economic theory that suggests stable long-run relationships. For example, $\boldsymbol{\beta} = (1, -1)'$ in the case of log stock price and log dividend.

Case 2. No, $\boldsymbol{\beta}$ must be estimated from the data. Assuming that $y_{it} \sim I(1)$, $i = 1, 2, \ldots, k$, and adopting the normalization scheme $\boldsymbol{\beta} = (1, -\boldsymbol{\beta}_2')'$, one can estimate $\boldsymbol{\beta}_2$ by the OLS regression:

$$y_{1t} = \beta_0 + \boldsymbol{\beta}_2'\mathbf{y}_{2t} + \varepsilon_t,$$

where $\mathbf{y}_{2t} = (y_{2t}, y_{3t}, \ldots, y_{kt})'$. The OLS estimates for β_0 and $\boldsymbol{\beta}_2$ are "super-consistent". Then one can apply the ADF or other unit-root tests to the residuals $\{\hat{\varepsilon}_t\}$, assuming no intercept and no linear trend. If the null hypothesis of a unit root is rejected, this provides evidence of cointegration. Note that the asymptotic distribution and critical values of the ADF test in this context differ from the standard case, due to the residuals being obtained from a regression involving I(1) variables.

- Pretesting the variables for their order of integration may be necessary. Two series of different orders of integration cannot be cointegrated.

- Shortcomings of the residual-based tests:
  - In small samples, the tests are not invariant to the choice of the dependent variable in the regression. (For example, residuals from regressing y_{1t} on $y_{2t}, y_{3t}, \ldots, y_{kt}$ differ from those obtained by regressing y_{2t} on $y_{1t}, y_{3t}, \ldots, y_{kt}$. In practice, one specification may suggest the presence of cointegration, while reversing the order may not.)
  - The tests do not allow for more than one cointegrating relationship in the case of multiple variables.
  - The tests generally have low statistical power.

3.3 Cointegrated VAR Models

- Consider the VAR(p) model:

$$\mathbf{y}_t = \mathbf{\Phi}_1 \mathbf{y}_{t-1} + \mathbf{\Phi}_2 \mathbf{y}_{t-2} + \ldots + \mathbf{\Phi}_p \mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t, \tag{16}$$

where $\boldsymbol{\varepsilon}_t \sim WN(\mathbf{0}, \mathbf{\Sigma})$, and the initial values $\mathbf{y}_0, \mathbf{y}_{-1}, \ldots, \mathbf{y}_{-p+1}$ are assumed to be given. Eqn. (16) can be rewritten as

$$\begin{aligned}\Delta \mathbf{y}_t + \mathbf{y}_{t-1} = \mathbf{\Phi}_1 \mathbf{y}_{t-1} + \mathbf{\Phi}_2 (\mathbf{y}_{t-1} - \Delta \mathbf{y}_{t-1}) + \ldots \\ + \mathbf{\Phi}_p (\mathbf{y}_{t-1} - \Delta \mathbf{y}_{t-1} - \ldots - \Delta \mathbf{y}_{t-p+1}) + \boldsymbol{\varepsilon}_t,\end{aligned}$$

or

$$\Delta \mathbf{y}_t = -\mathbf{\Pi} \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta \mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t, \tag{17}$$

where $\mathbf{\Pi} = \mathbf{I}_k - \mathbf{\Phi}_1 - \ldots - \mathbf{\Phi}_p$, and $\mathbf{\Gamma}_j = -\sum_{i=j+1}^{p} \mathbf{\Phi}_i$, for $j = 1, 2, \ldots, p-1$. (17) is known as the *vector error correction model* (VECM) of the VAR(p) process (16). $\mathbf{\Pi} \mathbf{y}_{t-1}$ is referred to as the *error correction term*.

- Recall that $\mathbf{\Phi}(L) = \mathbf{I}_k - \mathbf{\Phi}_1 L - \ldots - \mathbf{\Phi}_p L^p$. $\mathbf{y}_t$ is weakly stationary if all the roots of $|\mathbf{\Phi}(L)| = 0$ lie outside the unit circle. If $|\mathbf{\Phi}(1)| = 0$, then $\mathbf{y}_t$ is unit-root nonstationary.

- Suppose now $\mathbf{y}_t \sim I(1)$, then $|\mathbf{\Phi}(1)| = 0$, and $\mathbf{\Pi} = \mathbf{\Phi}(1)$ is singular, which implies that $\mathbf{\Pi}$ has reduced rank, i.e., Rank($\mathbf{\Pi}$) $< k$.

- Another way to see this result: Since $\mathbf{y}_t \sim I(1)$, then on the LHS of (17) we have $\Delta\mathbf{y}_t \sim I(0)$, and on the RHS both $\Delta\mathbf{y}_{t-j}$ and $\boldsymbol{\varepsilon}_t$ are $I(0)$. Note that $I(1) + I(0) = I(1)$. Consequently (17) holds if and only if $\mathbf{d}_t = \mathbf{\Pi}\mathbf{y}_{t-1} \sim I(0)$, which further implies that Rank($\mathbf{\Pi}$) $< k$ because otherwise $\mathbf{y}_{t-1} = \mathbf{\Pi}^{-1}\mathbf{d}_t \sim I(0)$, a contradiction to the assumption that $\mathbf{y}_t \sim I(1)$.

- Therefore, if $\mathbf{y}_{t-1} \sim I(1)$ and the linear combination $\mathbf{\Pi}\mathbf{y}_{t-1} \sim I(0)$ ($\mathbf{\Pi} \neq \mathbf{0}$), we say that *the VAR model (16) is cointegrated*. Rank($\mathbf{\Pi}$) plays a key role in the cointegration analysis. There are three cases:

Case 1. Rank($\mathbf{\Pi}$) $= 0$, which implies $\mathbf{\Pi} = \mathbf{0}$ and there are no cointegrating vectors. The VECM given by (17) reduces to a VAR($p-1$) in first differences:

$$\Delta\mathbf{y}_t = \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta\mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t.$$

$\Delta\mathbf{y}_t$ does not respond to the previous period's deviation from long-run equilibrium. In this case, we just run a normal VAR in $\Delta\mathbf{y}_t$ (first differences).

Case 2. Rank($\mathbf{\Pi}$) $= k$, i.e., $\mathbf{\Pi}$ has full rank, which implies that $|\mathbf{\Phi}(1)| \neq 0$, and $\mathbf{y}_t$ contains no unit roots. One should run a normal VAR in $\mathbf{y}_t$ (levels) directly.

Case 3. $0 <$Rank($\mathbf{\Pi}$) $= m < k$. In this case, we can write $\mathbf{\Pi}$ as

$$\underset{k\times k}{\mathbf{\Pi}} = \underset{(k\times m)}{\boldsymbol{\alpha}} \underset{(m\times k)}{\boldsymbol{\beta}'}, \tag{18}$$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are $k \times m$ matrices of full column ranks, i.e., Rank($\boldsymbol{\alpha}$) =Rank($\boldsymbol{\beta}$) $= m$. Then the VECM given by (17) can be rewritten as

$$\Delta\mathbf{y}_t = -\boldsymbol{\alpha}\boldsymbol{\beta}'\mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta\mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t. \tag{19}$$

Since $\mathbf{\Pi}\mathbf{y}_{t-1} = \boldsymbol{\alpha}\left(\boldsymbol{\beta}'\mathbf{y}_{t-1}\right) \sim I\left(0\right)$ and $\boldsymbol{\alpha}$ is full rank, we have

$$\boldsymbol{\beta}'\mathbf{y}_{t-1} \sim I\left(0\right),$$

where $\boldsymbol{\beta}'\mathbf{y}_t$ is the $m \times 1$ vector of *cointegrating relations*, or *long-run relations*; each column of $\boldsymbol{\beta}$ is a cointegrating vector; m (the rank of $\mathbf{\Pi}$) is the number of cointegrating relations, also known as the *dimension of the cointegration space*; and the elements of $\boldsymbol{\alpha}$ are known as the *adjustment parameters*. In this case, the VAR in levels is consistent but inefficient since it does not impose restrictions (or the cointegrating relations) on $\mathbf{\Phi}\left(1\right)$, whereas the VAR in differences is inconsistent since omitting the cointegrating relations entails a misspecification error.

- In sum, information about the long-run relationships among the variables is captured by the matrix $\mathbf{\Pi}$. Cointegration analysis centers on analyzing the structure of $\mathbf{\Pi}$.
- The VECM representation (19) has nice interpretations. As an illustration, consider the following example.

Example. Bivariate VECM for short-term and long-term interest rates.
Let r_{St} and r_{Lt} denote the short-term and long-term interest rates, respectively, and assume that both interest rates are $I\left(1\right)$. The model is

$$\begin{aligned}
\Delta r_{St} &= \alpha_S\left(r_{Lt-1} - \beta r_{St-1}\right) + \varepsilon_{St}, \quad \alpha_S > 0, \\
\Delta r_{Lt} &= -\alpha_L\left(r_{Lt-1} - \beta r_{St-1}\right) + \varepsilon_{Lt}, \quad \alpha_L > 0,
\end{aligned}$$

where ε_{St} and ε_{Lt} are white noise errors that may be correlated.
We make the following observations:

1. By assumption, $\Delta r_{St}, \Delta r_{Lt}, \varepsilon_{St}, \varepsilon_{Lt}$ are all $I\left(0\right)$, and therefore the linear combination $r_{Lt-1} - \beta r_{St-1}$ must be $I\left(0\right)$. That is, the two rates are cointegrated with vector $\left(1, -\beta\right)'$.
2. The terms in parentheses are deviations from the long-run equilibrium, which is reached if $r_{Lt-1} = \beta r_{St-1}$ so that the expected change in each rate is zero. If deviations are positive,

namely, $r_{Lt-1} - \beta r_{St-1} > 0$, r_{St} would rise and r_{Lt} would fall. $u_{t-1} \equiv r_{Lt-1} - \beta r_{St-1}$ is referred to as the (lagged) *disequilibrium error* or *cointegrating residual*.

3. α_S and α_L measure the speed of adjustment. The larger the values, the greater the responses of r_{St} and r_{Lt} to the last period's deviation from long-run equilibrium.

4. The model can be extended to

$$\begin{aligned}
\Delta r_{St} &= \alpha_S \left(r_{Lt-1} - \beta r_{St-1}\right) + \gamma_{11}\Delta r_{St-1} + \gamma_{12}\Delta r_{Lt-1} + \varepsilon_{St}, \quad \alpha_S > 0, \\
\Delta r_{Lt} &= -\alpha_L \left(r_{Lt-1} - \beta r_{St-1}\right) + \gamma_{21}\Delta r_{St-1} + \gamma_{22}\Delta r_{Lt-1} + \varepsilon_{Lt}, \quad \alpha_L > 0,
\end{aligned}$$

and higher order lags of differences can also be included. The above discussions remain valid.

3.4 Identification of Long-Run Effects

- In general, the factorization in (18) is not unique. To see this, consider any $m \times m$ nonsingular matrix $\mathbf{Q}$,

$$\boldsymbol{\alpha}\boldsymbol{\beta}' = \boldsymbol{\alpha}\mathbf{Q}\mathbf{Q}^{-1}\boldsymbol{\beta}' = (\boldsymbol{\alpha}\mathbf{Q})\left(\boldsymbol{\beta}\mathbf{Q}^{-1\prime}\right)' \equiv \tilde{\boldsymbol{\alpha}}\tilde{\boldsymbol{\beta}}',$$

where both $\tilde{\boldsymbol{\alpha}}$ and $\tilde{\boldsymbol{\beta}}$ are also of rank m. Therefore, the factorization $\boldsymbol{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}'$ only identifies the space spanned by the cointegrating relations. If $\boldsymbol{\beta}'\mathbf{y}_{t-1} \sim I(0)$, so will $\tilde{\boldsymbol{\beta}}'\mathbf{y}_{t-1} \sim I(0)$. The m columns of $\boldsymbol{\beta}$ and $\tilde{\boldsymbol{\beta}}$ are both valid as cointegrating vectors. This is known as the *long-run identification problem*.

- Additional constraints are needed in order to uniquely identify $\boldsymbol{\beta}$. To exactly identify the cointegrating coefficients, we need m^2 exact- or just-identifying restrictions—m restrictions on each of the m cointegrating relations. A common normalization scheme is to set $\boldsymbol{\beta} = (\mathbf{I}_m, \boldsymbol{\beta}_1')'$, where $\boldsymbol{\beta}_1$ is a $(k-m) \times m$ matrix.

- For example, consider a bivariate cointegrated VAR model, where there exists a 2×1

vector $\boldsymbol{\beta} = (\beta_1, \beta_2)'$ such that

$$\boldsymbol{\beta}'\mathbf{y}_t = \beta_1 y_{1t} + \beta_2 y_{2t} \sim I(0).$$

Under the normalization $\beta_1 = 1$ and $\beta_2 = -\beta$, the cointegrating relation becomes $\boldsymbol{\beta}'\mathbf{y}_t = y_{1t} - \beta y_{2t}$, which implies the long-run equilibrium relation $y_{1t} = \beta y_{2t}$. One can find the value of β from $\boldsymbol{\Pi}$ by matching the coefficients:

$$\boldsymbol{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}' = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \begin{pmatrix} 1 & -\beta \end{pmatrix} = \begin{pmatrix} \alpha_1 & -\alpha_1\beta \\ \alpha_2 & -\alpha_2\beta \end{pmatrix} = \begin{pmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{pmatrix}.$$

3.5 Modeling Cointegration

3.5.1 Basic steps

- Basic steps in Johansen's methodology (1988, 1991):
  - Specify and estimate an unrestricted VAR(p) model for $\mathbf{y}_t$, using the *undifferenced* data, which may involve potentially non-stationary variables.
  - Perform the likelihood ratio tests for the rank of $\boldsymbol{\Pi}$ to determine the number of cointegrating vectors.
  - Impose appropriate normalization and identifying restrictions on the cointegrating vectors.
  - Estimate the resulting cointegrated VECM using maximum likelihood.

3.5.2 Tests for cointegration in VAR models

- Johansen (1988, 1991) proposes two different likelihood ratio (LR) tests: the trace test and the maximum eigenvalue test. They are essentially tests on the rank of the matrix $\boldsymbol{\Pi}$.
- Unlike the residual-based cointegration tests, Johansen's cointegration tests allow for multiple cointegrating relations, and the test statistics are invariant to the ordering of variables in $\mathbf{y}_t$.

- **Trace cointegration test:** The hypotheses of interest are

$$H_0(m) : \text{Rank}(\mathbf{\Pi}) = m, \quad \text{vs.} \quad H_1(k) : m < \text{Rank}(\mathbf{\Pi}) \leq k,$$

for $m = 0, 1, 2, \ldots, k-1$. The LR statistic, called the *trace statistic*, is given by

$$LR_{tr}(m) = -2\left[\ell^c\left(\hat{\boldsymbol{\beta}}; m\right) - \ell^c\left(\hat{\boldsymbol{\beta}}; k\right)\right] = -T\sum_{i=m+1}^{k}\log\left(1-\hat{\lambda}_i\right),$$

where $\ell^c\left(\hat{\boldsymbol{\beta}}; m\right)$ and $\ell^c\left(\hat{\boldsymbol{\beta}}; k\right)$ denote the (concentrated) maximized log-likelihood under $H_0(m)$ and $H_1(k)$, respectively,[2] and $\hat{\lambda}_1 > \hat{\lambda}_2 > \ldots > \hat{\lambda}_k$ are the estimated eigenvalues of matrix $\mathbf{\Pi}$. These eigenvalues also equal the squared *canonical correlations* between $\Delta\mathbf{y}_t$ and $\mathbf{y}_{t-1}$ corrected for lagged $\Delta\mathbf{y}_t$ (and other determinants) and hence they lie between 0 and 1.

- Main idea of the test: Recall that $\text{Rank}(\mathbf{\Pi})$ equals the number of nonzero eigenvalues of $\mathbf{\Pi}$. If $\text{Rank}(\mathbf{\Pi}) = m$, then $\hat{\lambda}_i$ (and $\log\left(1-\hat{\lambda}_i\right)$) should be small for $i > m$. The $LR_{tr}(m)$ statistic depends on the smallest $k-m$ eigenvalues and hence should be small under the null. If $\text{Rank}(\mathbf{\Pi}) > m$, then some of the $\hat{\lambda}_i$ for $i > m$ will be nonzero (but less than one), and hence $\log\left(1-\hat{\lambda}_i\right) << 0$ for some $i > m$, resulting in a large $LR_{tr}(m)$.

- **Maximum eigenvalue cointegration test:** The hypotheses of interest are

$$H_0(m) : \text{Rank}(\mathbf{\Pi}) = m, \quad \text{vs.} \quad H_1(m+1) : \text{Rank}(\mathbf{\Pi}) = m+1,$$

for $m = 0, 1, 2, \ldots, k-1$. The LR statistic, called the *maximum eigenvalue statistic*, is given by

$$LR_{max}(m) = -2\left[\ell^c\left(\hat{\boldsymbol{\beta}}; m\right) - \ell^c\left(\hat{\boldsymbol{\beta}}; m+1\right)\right] = -T\log\left(1-\hat{\lambda}_{m+1}\right).$$

- Because of the existence of unit roots, both the trace and maximum eigenvalue test statis-

[2] The parameters in $\boldsymbol{\alpha}$ and $\mathbf{\Sigma}$ have been concentrated out. That is, we first solve for $\hat{\boldsymbol{\alpha}}(\boldsymbol{\beta})$ and $\hat{\mathbf{\Sigma}}(\boldsymbol{\beta})$ as explicit functions of $\boldsymbol{\beta}$, and then replace $\boldsymbol{\alpha}$ and $\mathbf{\Sigma}$ in the log-likelihood function. This yields the "concentrated" log-likelihood, which depends only on $\boldsymbol{\beta}$.

tics do not follow a standard limiting χ^2-distribution. The critical values are obtained by simulations.

- To find the number of cointegrating relations, the strategy is to test a sequence of null hypotheses

$$H_0(0) : \text{Rank}(\mathbf{\Pi}) = 0, \quad H_0(1) : \text{Rank}(\mathbf{\Pi}) = 1, \ldots, \quad H_0(k-1) : \text{Rank}(\mathbf{\Pi}) = k-1,$$

  and terminate the tests when an H_0 cannot be rejected for the first time. The cointegrating rank is then selected accordingly. Both the trace and maximum eigenvalue tests can be used in this sequential procedure.

- In finite samples, the trace test tends to have more distorted sizes, but it is usually more robust and holds more power since it considers all of the smallest eigenvalues (Cheung and Lai, 1993; Lutkepohl et al., 2001).

- Since the eigenvectors are linear combinations of $\mathbf{y}_t$, they provide an estimate of the cointegrating vectors. For a given $0 <\text{Rank}(\mathbf{\Pi}) = m < k$, an estimate of $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{v}}_1, \hat{\boldsymbol{v}}_2, \ldots, \hat{\boldsymbol{v}}_m)$, where $\hat{\boldsymbol{v}}_i$ are the eigenvectors associated with the eigenvalues $\hat{\lambda}_i$ (more on estimation below).

3.5.3 Specification of the deterministic terms

- As with the univariate case, the asymptotic distribution (and hence the critical values) of the cointegration tests depends on the deterministic terms. Johansen's tests can be viewed as a multivariate generalization of the Dickey-Fuller test.

- Consider the following general VEC($p-1$) model:

$$\Delta \mathbf{y}_t = \boldsymbol{a}_0 + \boldsymbol{a}_1 t - \boldsymbol{\alpha}\boldsymbol{\beta}' \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta \mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t. \tag{20}$$

  There are five cases:

Case 1. (no intercepts and no trends) $\boldsymbol{a}_0 = \boldsymbol{a}_1 = \mathbf{0}$. All components series of $\mathbf{y}_t$ are $I(1)$ without drift and the cointegrating relations $\boldsymbol{\beta}'\mathbf{y}_t$ have zero means.

$$\Delta\mathbf{y}_t = -\boldsymbol{\alpha}\boldsymbol{\beta}'\mathbf{y}_{t-1} + \sum_{j=1}^{p-1}\boldsymbol{\Gamma}_j\Delta\mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t.$$

Case 2. (restricted intercepts and no trends) $\boldsymbol{a}_0 = \boldsymbol{\alpha}\boldsymbol{c}_0$ and $\boldsymbol{a}_1 = \mathbf{0}$, where $\boldsymbol{c}_0$ is an $m \times 1$ nonzero constant vector. In this case, equation (20) becomes

$$\Delta\mathbf{y}_t = -\boldsymbol{\alpha}\left(\boldsymbol{\beta}'\mathbf{y}_{t-1} - \boldsymbol{c}_0\right) + \sum_{j=1}^{p-1}\boldsymbol{\Gamma}_j\Delta\mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t,$$

so that the components series of $\mathbf{y}_t$ are $I(1)$ without drift and the cointegrating relations $\boldsymbol{\beta}'\mathbf{y}_t$ have a nonzero mean vector, $\boldsymbol{c}_0$.

Case 3. (unrestricted intercepts and no trends) $\boldsymbol{a}_0 \neq \mathbf{0}$ and $\boldsymbol{a}_1 = \mathbf{0}$. The components series of $\mathbf{y}_t$ are $I(1)$ with a drift vector $\boldsymbol{a}_0$ and the cointegrating relations $\boldsymbol{\beta}'\mathbf{y}_t$ may have a nonzero mean.

$$\Delta\mathbf{y}_t = \boldsymbol{a}_0 - \boldsymbol{\alpha}\boldsymbol{\beta}'\mathbf{y}_{t-1} + \sum_{j=1}^{p-1}\boldsymbol{\Gamma}_j\Delta\mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t.$$

Case 4. (unrestricted intercepts and restricted trends) $\boldsymbol{a}_0 \neq \mathbf{0}$ and $\boldsymbol{a}_1 = \boldsymbol{\alpha}\boldsymbol{c}_1$, where $\boldsymbol{c}_1$ is an $m \times 1$ nonzero constant vector. In this case, (20) becomes

$$\Delta\mathbf{y}_t = \boldsymbol{a}_0 - \boldsymbol{\alpha}\left(\boldsymbol{\beta}'\mathbf{y}_{t-1} - \boldsymbol{c}_1 t\right) + \sum_{j=1}^{p-1}\boldsymbol{\Gamma}_j\Delta\mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t,$$

so that the components series of $\mathbf{y}_t$ are $I(1)$ with a drift vector $\boldsymbol{a}_0$ and the cointegrating relations $\boldsymbol{\beta}'\mathbf{y}_t$ have a linear trend term, $\boldsymbol{c}_1 t$.

Case 5. (unrestricted intercepts and trends) $\boldsymbol{a}_0 \neq \mathbf{0}$ and $\boldsymbol{a}_1 \neq \mathbf{0}$. The components series of $\mathbf{y}_t$ are $I(1)$ with a quadratic time trend and the cointegrating relations $\boldsymbol{\beta}'\mathbf{y}_t$ have a linear trend.

$$\Delta\mathbf{y}_t = \boldsymbol{a}_0 + \boldsymbol{a}_1 t - \boldsymbol{\alpha}\boldsymbol{\beta}'\mathbf{y}_{t-1} + \sum_{j=1}^{p-1}\boldsymbol{\Gamma}_j\Delta\mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t.$$